



RURO Post-Estimation Analysis

Generated: 2026-05-13 18:39:01

Total Parameters: 48 | **Groups:** joint

Estimation Results Source:

U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0a\run_2026-05-13_18-28-19\estimation_results.json

Estimation Configuration

Specification	ruro_occ_M0a
Specification File	scripts\enhanced\estimation_spec_ruro_occ_M0a.yaml
Model Family	regular
Proposal Correction	Enabled
Proposal Correction Form	-log(prior)
Opportunity Centering	Enabled

Single Males Single Females Males in Couples Females in Couples

 Elapsed Time

Estimation	Post-Estimation	Total	Iterations
4m 44s	20.7s	5m 5s	27



Model Fit Statistics

log_likelihood	-6521.4292
n_observations	4.2530e+05

n_groups	4253.0000
n_parameters	48.0000
AIC	1.3139e+04
BIC	1.3665e+04
ll_null	-1.9586e+04
ll_null_uniform	-1.9586e+04
ll_null_prior_corrected	-2.2322e+04
n_obs_long	4.2530e+05
rho_squared_uniform	0.6670
rho_squared_adj_uniform	0.6646
rho_squared_prior_corrected	0.7078
rho_squared_adj_prior_corrected	0.7057
rho_squared	0.7078
rho_squared_adj	0.7057
AIC_per_obs	0.0309
n_negative_muc_total	0.0000e+00
n_negative_mul_total	0.0000e+00
pct_negative_muc_total	0.0000e+00
pct_negative_mul_total	0.0000e+00
n_bounded_params	48
n_hit_lower_bound	0
n_hit_upper_bound	0



Understanding Bounded Parameters

- **n_bounded_params (48):** Parameters with constraints defined.
- **n_hit_lower_bound (0):** Parameters at lower bound. ✓ Good!

- **n_hit_upper_bound (0):** Parameters at upper bound. ✓ Good!

 Identification Diagnostics

Hessian-based diagnostics for parameter identification quality.

Condition Number: 1.06e+10

✗ Severe Identification Problem

The Hessian is severely ill-conditioned ($\kappa \geq 10,000$). This indicates serious identification issues with unreliable parameter estimates.

Eigenvalue Analysis

Minimum Eigenvalue

-3.81e+00

Maximum Eigenvalue

1.50e+10

Negative Eigenvalues

1

✗ NOT a local maximum!

Eigenvalue Summary

p5 / median / p95: 1.75e+00 / 1.97e+02 / 7.81e+05

Smallest 5: -3.81e+00, 1.41e+00, 1.63e+00, 1.97e+00, 2.87e+00

Largest 5: 4.98e+05, 5.56e+05, 9.03e+05, 1.08e+06, 1.50e+10

High Correlations ($|\rho| \geq 0.90$)

Parameter A	Parameter B	Correlation
beta_c_sm	beta_c_sf	-4.673
beta_c_sm	theta_c_sf	-3.794
theta_c_sm	beta_c_sf	-3.713
theta_c_sm	theta_c_sf	-3.010

Parameter A	Parameter B	Correlation
beta_c_sm	beta_c	-1.894
beta_c_sm	sigma	+1.635
theta_c_sm	beta_c	-1.486
beta_c_sf	beta_c	-1.387
beta_c_sf	sigma	+1.343
theta_c_sm	sigma	+1.271
beta_l0_sf	beta_c_sf	-1.236
theta_c_sf	beta_c	-1.099
theta_c_sf	sigma	+1.053
beta_c_sm	theta_l_sf	-1.041
beta_l0_sf	theta_c_sf	-1.018
beta_c	theta_c	+1.011
beta_c_sm	theta_c_sm	+0.991
beta_c_sf	theta_c_sf	+0.990
beta_w_pexp	beta_w_pexp2	-0.962
beta_c_sm	theta_c	-0.956

 Technical Notes

Condition Number (κ): Measures sensitivity of parameter estimates to small changes in data. $\kappa = \lambda_{\max} / \lambda_{\min}$, where λ are eigenvalues of $-H$ (Hessian).

Eigenvalues: All eigenvalues of $-H$ should be positive for a local maximum. Near-zero eigenvalues indicate flat directions (weak identification).

Sign convention: Eigenvalues shown come from the Hessian used to compute SEs. When SEs are computed numerically, this is the Hessian of the negative log-likelihood (H), so negative values indicate H is not positive semidefinite (ill-conditioning or non-optimum).

Standard Errors: Computed as $SE = \sqrt{\text{diag}((-H)^{-1})}$, where H is the Hessian at the optimum. Large SE indicates parameter is not precisely estimated.



Identification Diagnostics (Estimation Stage)

Source:

U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0a\run_2026-05-13_18-28-19\identification_diagnostics.txt

```
Identification Diagnostics
=====
specification: ruro_occ_M0a
n_parameters: 48
bounded_hits_total: 0
bounded_hits_lower: 0
bounded_hits_upper: 0
hessian_condition_number: 10648346567.961565
hessian_negative_eigenvalues: 1
hessian_near_zero_eigenvalues(|eig|<=1e-8): 0
negative_variances_from_varcov: 0

Parameter Stability
-----
reference: initial_vector
matched_params: 48/48
delta_l2: 9.707291e+00
delta_l2_relative: 2.269174e+00
delta_max_abs: 5.153684e+00
delta_mean_abs: 7.916560e-01

Bound Hits
-----
none
```



Model Specification

Utility and opportunity functions for each demographic group, shown in both symbolic and numerical form.

Single Males

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{I0} + \beta_{I,age} \cdot age + \beta_{I,age2} \cdot age2$$

Utility Function (Numerical)

$$U = 0.7475 \cdot BC(C, -0.8357) + (3.9530 + 0.0099 \cdot age + 0.0020 \cdot age2) \cdot BC(L, -0.7073)$$

Single Females

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{I0} + \beta_{I,age} \cdot age + \beta_{I,age2} \cdot age2 + \beta_{I,nkids} \cdot nkids$$

Utility Function (Numerical)

$$U = 0.6767 \cdot BC(C, -0.8357) + (4.5315 + 0.0027 \cdot age + 0.0042 \cdot age2 + 0.0949 \cdot nkids) \cdot BC(L, -0.7268)$$

Males in Couples

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{10} + \beta_{1,age} \cdot age + \beta_{1,age2} \cdot age2$$

Utility Function (Numerical)

$$U = 6.1537 \cdot BC(C, 0.3189) + (2.3263 - 0.0044 \cdot age + 0.0015 \cdot age2) \cdot BC(L, -0.7392)$$

Females in Couples

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{10} + \beta_{1,age} \cdot age + \beta_{1,age2} \cdot age2 + \beta_{1,nkids} \cdot nkids$$

Utility Function (Numerical)

$$U = 6.1537 \cdot BC(C, 0.3189) + (4.8450 - 0.0467 \cdot age + 0.0024 \cdot age2 + 0.2412 \cdot nkids) \cdot BC(L, -0.6970)$$

Model Index

$$V_{ij} = U_{ij} + O^E_{ij} + O^H_{ij} + O^W_{ij} + O^{Occ}_{ij} + \log \text{prior}_{ij}$$

$$P_{ij} = \exp(V_{ij}) / \sum_{k \in C_i} \exp(V_{ik})$$

U is the utility of consumption and leisure; the O' terms are opportunity-density contributions (employment/hours, wage, occupation); log prior rescales each alternative to the data-generating measure. The terms displayed reflect the opportunity blocks declared in the active YAML specification.



Employment and Hours Opportunity Parameters

Working-gated employment and hours intercepts. Combines hours_opportunity.shifters (base employment intercept and PT/FT focal-point indicators) with market_opportunity.shifters (working-gated residual market access terms like gsur, education).

Symbolic form

$$O^E + O^H = \text{beta_E} * \text{working} + \text{beta_h_pt1} * \text{working_pt1} + \text{beta_h_pt2} * \text{working_pt2} + \text{beta_h_ft} * \text{working_ft} + \text{beta_E_gsur} * \text{gsur} * \text{working} + \text{beta_E_educH} * \text{educH} * \text{working}$$

Numerical form (by group)

Joint (All Groups): $-2.7597 * \text{working} + -0.4991 * \text{working_pt1} + 0.3772 * \text{working_pt2} + 1.4536 * \text{working_ft} + -0.7408 * \text{gsur} *$


```
working + 0.5495 * educH * working
```



Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\log(\text{wage}) = \mu_w(X) + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. The mean μ_w is built from `wage_opportunity.mean_shifters`.

Symbolic form

$$\begin{aligned} \mu_w = & \\ & \text{beta_w0} \\ & + \text{beta_w_educL} \cdot \text{educL} \\ & + \text{beta_w_educH} \cdot \text{educH} \\ & + \text{beta_w_pexp} \cdot \text{pexp_years} \\ & + \text{beta_w_pexp2} \cdot \text{pexp_years2} \end{aligned}$$

$$\log(\text{wage}) = \mu_w + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Numerical form (by group)

Joint (All Groups): $\mu_w = 2.0476 + -0.0478 * \text{educL} + 0.3007 * \text{educH}$
 $+ 0.0167 * \text{pexp_years} + -0.0002 * \text{pexp_years2}$
 $\sigma = 0.4189$



Occupation Opportunity Parameters

Working-gated occupation shifters. Each `applies_to` group has its own coefficient set on the non-reference categories of `loc4`. Reference category: `loc4 = 1`.

Single Males (`applies_to = sm`)

$$\begin{aligned}
 O_{sm}^{Occ} = & \\
 & \beta_{occ_2_sm} * loc4_2 * working \\
 & + \beta_{occ_3_sm} * loc4_3 * working \\
 & + \beta_{occ_4_sm} * loc4_4 * working \\
 & -1.5154 * loc4_2 * working + -2.1726 * loc4_3 * \\
 & working + 0.0169 * loc4_4 * working
 \end{aligned}$$

Single Females (`applies_to = sf`)

$$\begin{aligned}
 O_{sf}^{Occ} = & \\
 & \beta_{occ_2_sf} * loc4_2 * working \\
 & + \beta_{occ_3_sf} * loc4_3 * working \\
 & + \beta_{occ_4_sf} * loc4_4 * working \\
 & -0.0211 * loc4_2 * working + -0.5724 * loc4_3 * \\
 & working + 0.7876 * loc4_4 * working
 \end{aligned}$$

couples_male (`applies_to = cm`)

$$\begin{aligned}
 O_{cm}^{Occ} = & \\
 & \beta_{occ_2_cm} * loc4_2 * working \\
 & + \beta_{occ_3_cm} * loc4_3 * working
 \end{aligned}$$

```
+ beta_occ_4_cm * loc4_4 * working
```

```

-1.4730 * loc4_2 * working + -2.2112 * loc4_3 *
working + 0.4791 * loc4_4 * working

```

couples_female (applies_to = cf)

$$O_{cf}^{Occ} =$$

```

beta_occ_2_cf * loc4_2 * working
+ beta_occ_3_cf * loc4_3 * working
+ beta_occ_4_cf * loc4_4 * working

```

```

0.1878 * loc4_2 * working + -0.2020 * loc4_3 *
working + 1.1237 * loc4_4 * working

```



Curvature-Based Heuristics (Structural Elasticity Approximations)



Interpretation Note

These are **not** true labor supply elasticities. They are heuristic approximations derived from the curvature parameters (θ) of the Box-Cox utility function. The Hicksian approximation is $(1 - \theta_l)$, which measures the curvature of preferences. For rigorous elasticity estimates, use simulation-based methods that account for the full discrete choice structure and budget constraints.

 Labor Supply Elasticities

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Males	1.707	1.607	0.512	1.195	-0.707	-0.836	3.953	0.748
Single Females	1.727	1.627	0.518	1.209	-0.727	-0.836	4.532	0.677
Males in Couples	1.739	1.639	0.522	1.217	-0.739	0.319	2.326	6.154
Females in Couples	1.697	1.597	0.509	1.188	-0.697	0.319	4.845	6.154

 Fit Diagnostics

Observed vs Predicted

Group	Obs. Participation	Pred. Participation	Obs. Mean Hours	Pred. Mean Hours
Single Males	94.0%	100.0%	36.3	34.4
Single Females	93.0%	100.0%	39.3	34.4
Males in Couples	97.2%	100.0%	41.6	58.7
Females in Couples	96.5%	100.0%	35.6	58.5

 Marginal Utility Diagnostics

Analysis at chosen alternatives. Negative values indicate potential specification issues.

Negative MUC/MUL Summary

MUC Behavior Analysis

For well-behaved utility: $MUC > 0$ ($\beta_c > 0$) and diminishing ($\theta_c < 1$)

Group	N Individuals	% Neg MUC	% Neg MUL	MUC Mean	MUL Mean
Single Males	910	0.00%	0.00%	0.0000	7.1592e-03
Single Females	766	0.00%	0.00%	0.0000	8.9421e-03
Males in Couples	2577	0.00%	0.00%	0.0218	5.4153e-03
Females in Couples	2577	0.00%	0.00%	0.0218	8.8440e-03

Group	β_c	θ_c	MUC Positive?	MUC Diminishing?	Well-Behaved?	MUC at Median C	C where MUC=1	Notes
Single Males	0.7475	-0.8357	✓	✓	✓	0.7475	0.8534	
Single Females	0.6767	-0.8357	✓	✓	✓	0.6767	0.8084	
Males in Couples	6.1537	0.3189	✓	✓	✓	6.1537	14.4078	
Females in Couples	6.1537	0.3189	✓	✓	✓	6.1537	14.4078	

 Choice Probability Diagnostics

Probability Sum Sanity Check

Max $ \sum p - 1 $	0.000000
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Mean $ \Sigma p - 1 $	0.000000
% HH off by >0.01	0.00%
% HH off by >0.001	0.00%

P(chosen) Distribution

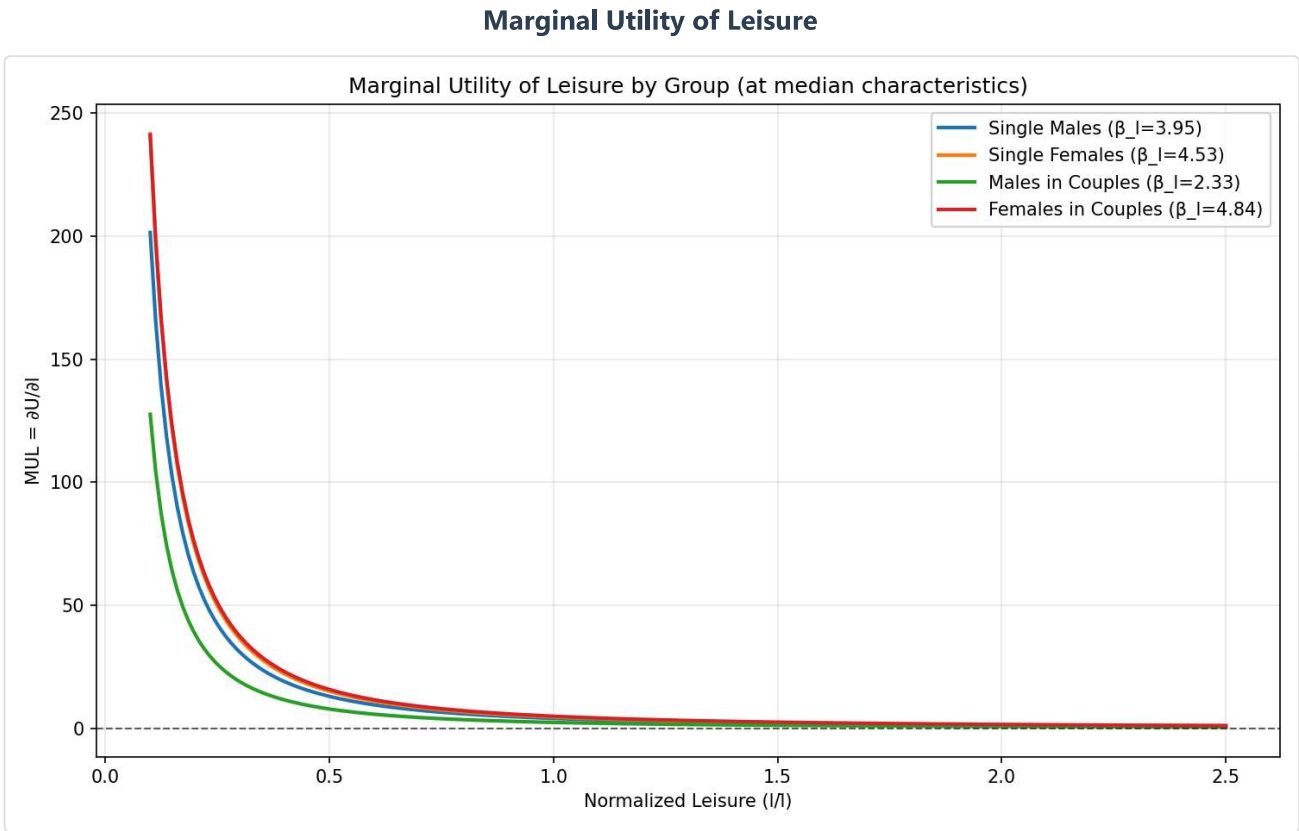
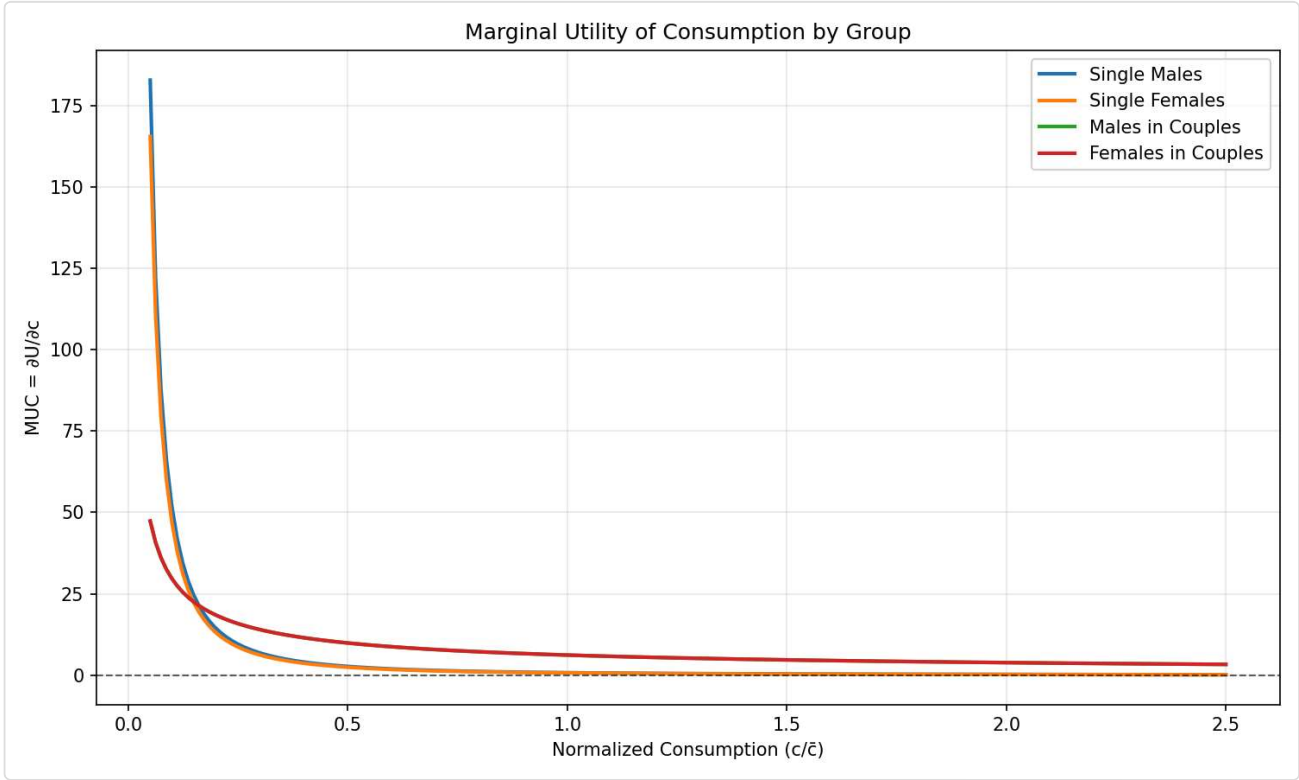
Min	0.0000
10th percentile	0.0126
25th percentile	0.0712
Median	0.2526
Mean	0.3679
75th percentile	0.6701
90th percentile	0.9063
Max	0.9997



Utility Contours & Plots

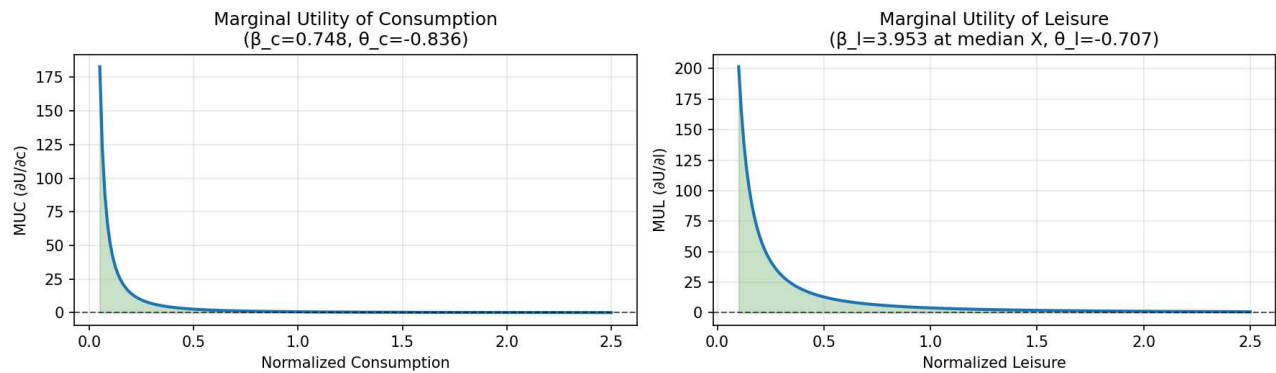
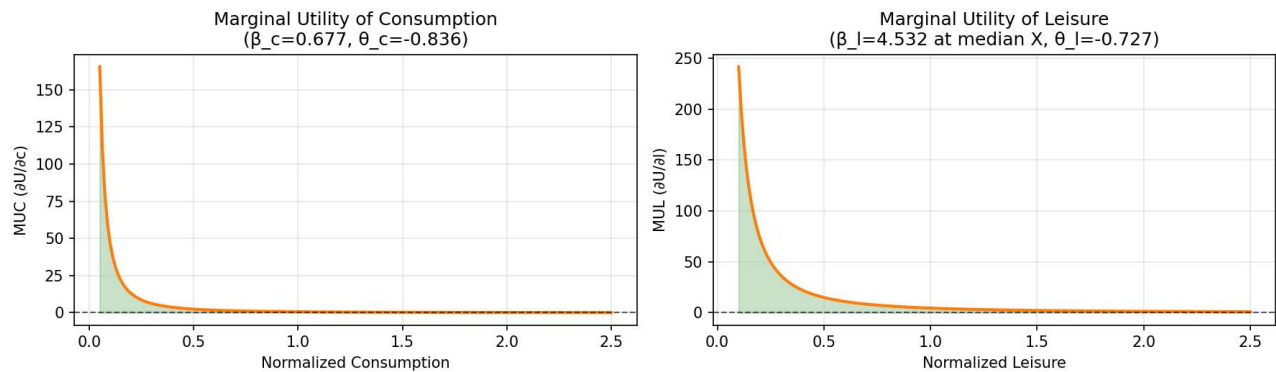
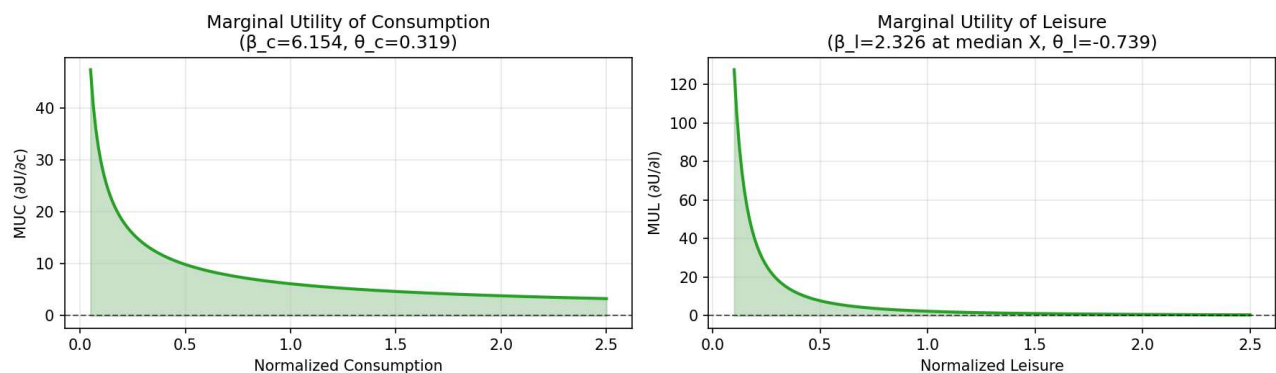
Marginal Utility Comparison

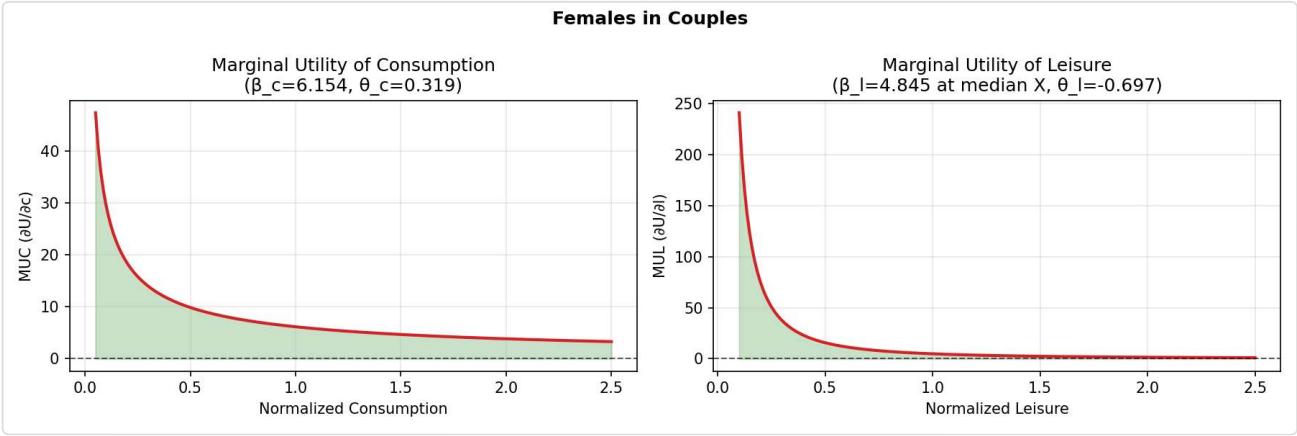
Marginal Utility of Consumption



Marginal Utility Distributions by Group

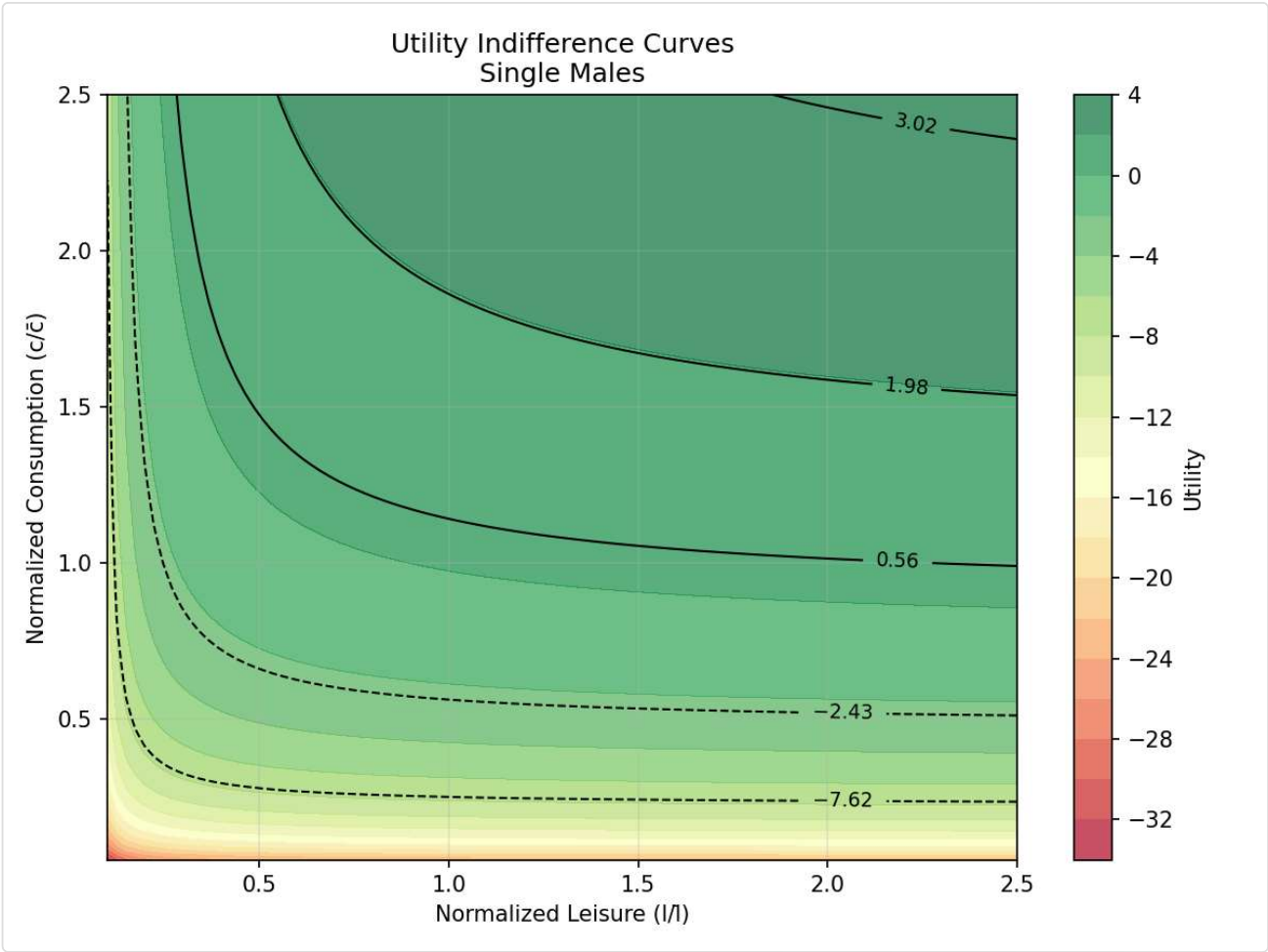
Single Males - MUC & MUL Distributions

Single Males**Single Females - MUC & MUL Distributions****Single Females****Males in Couples - MUC & MUL Distributions****Males in Couples****Females in Couples - MUC & MUL Distributions**

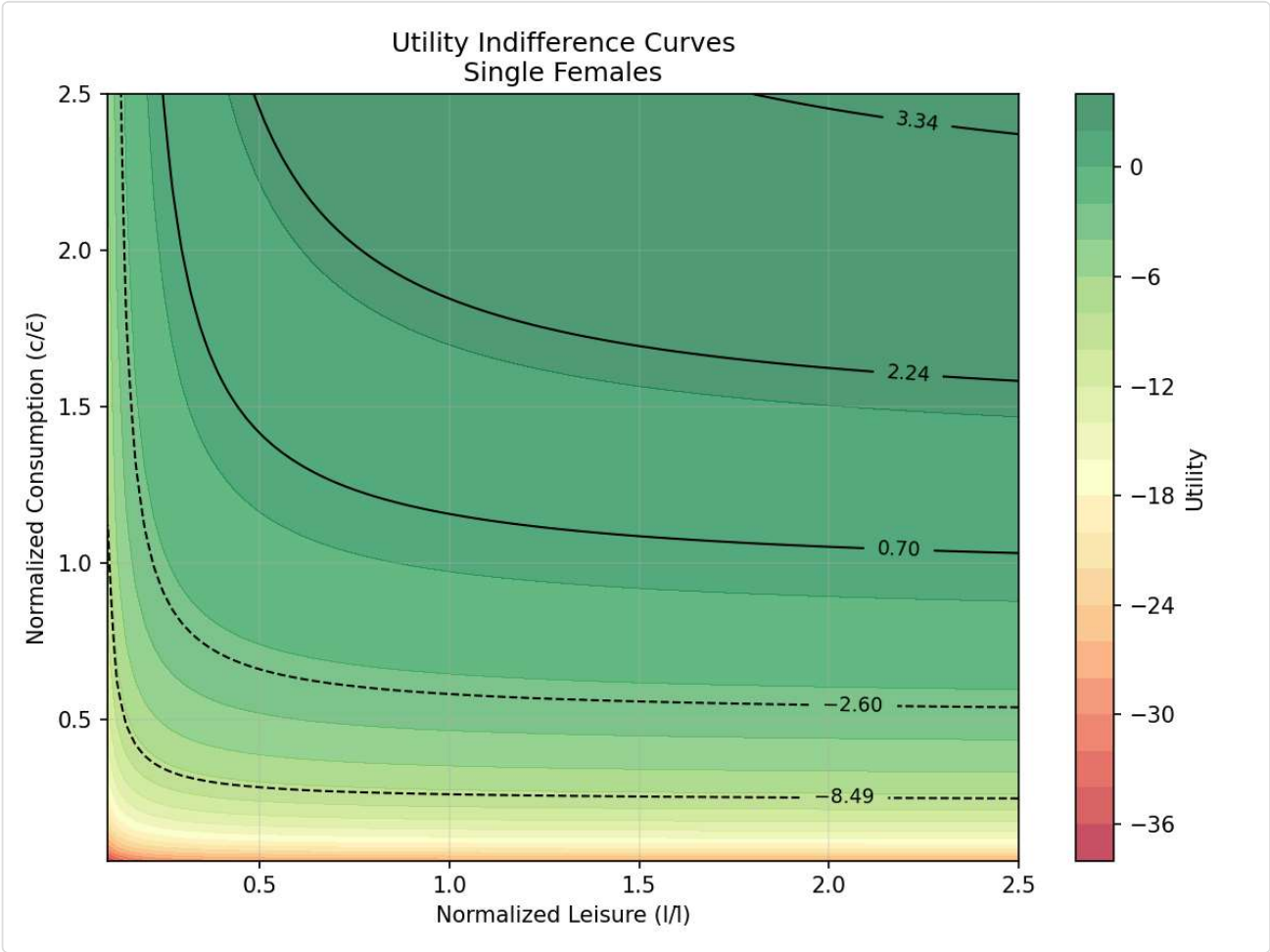


Utility Indifference Curves by Group

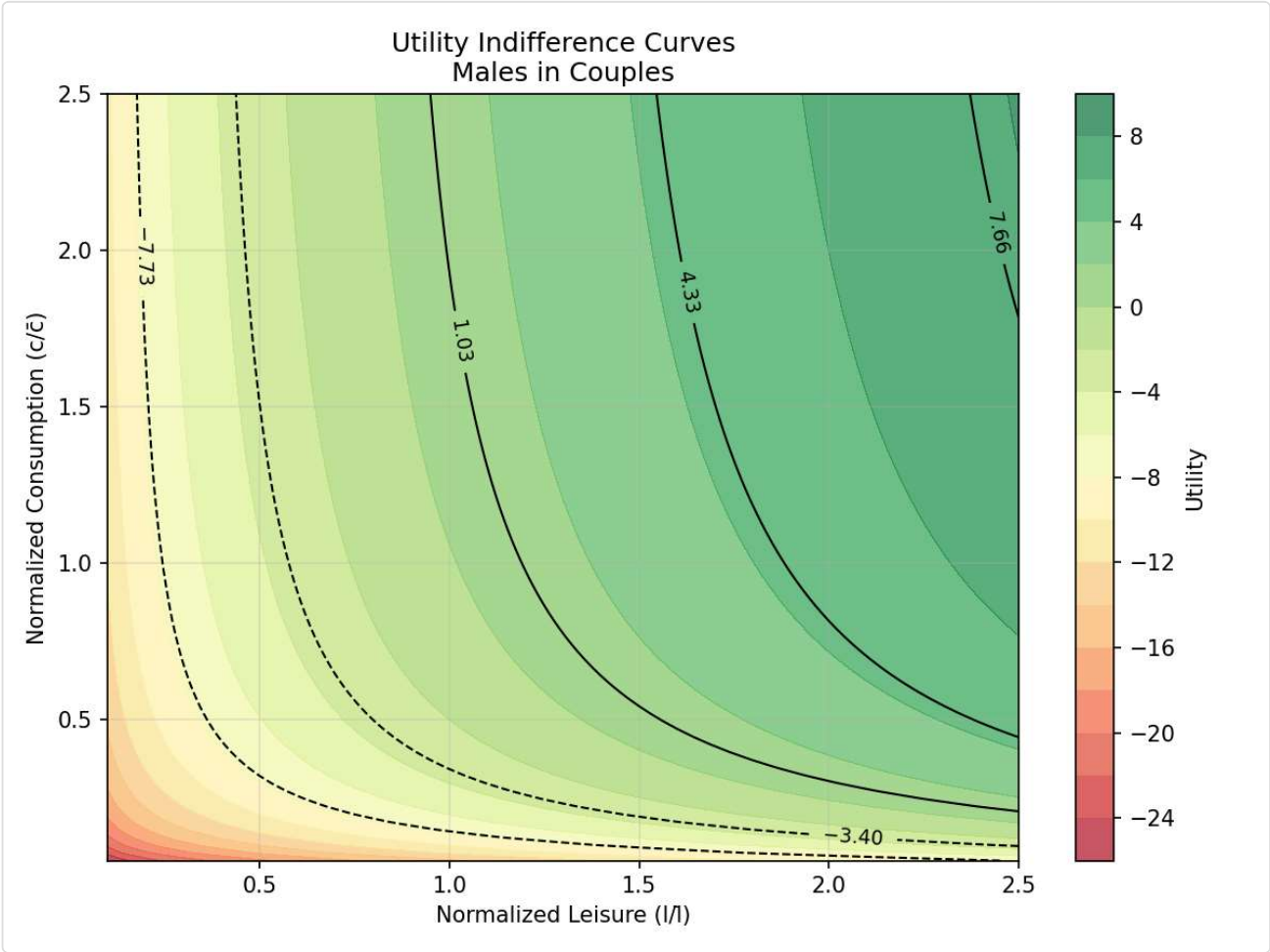
Single Males



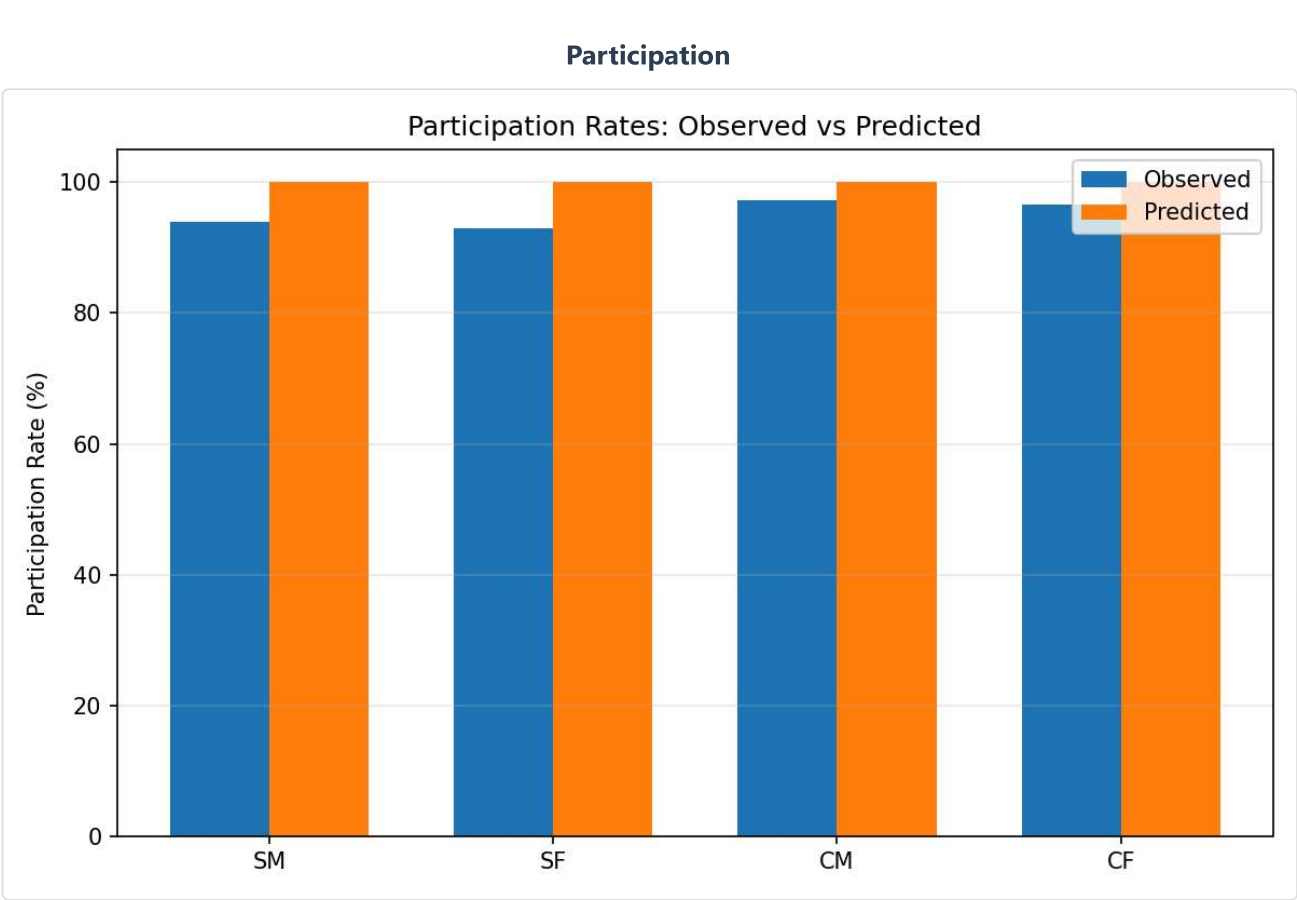
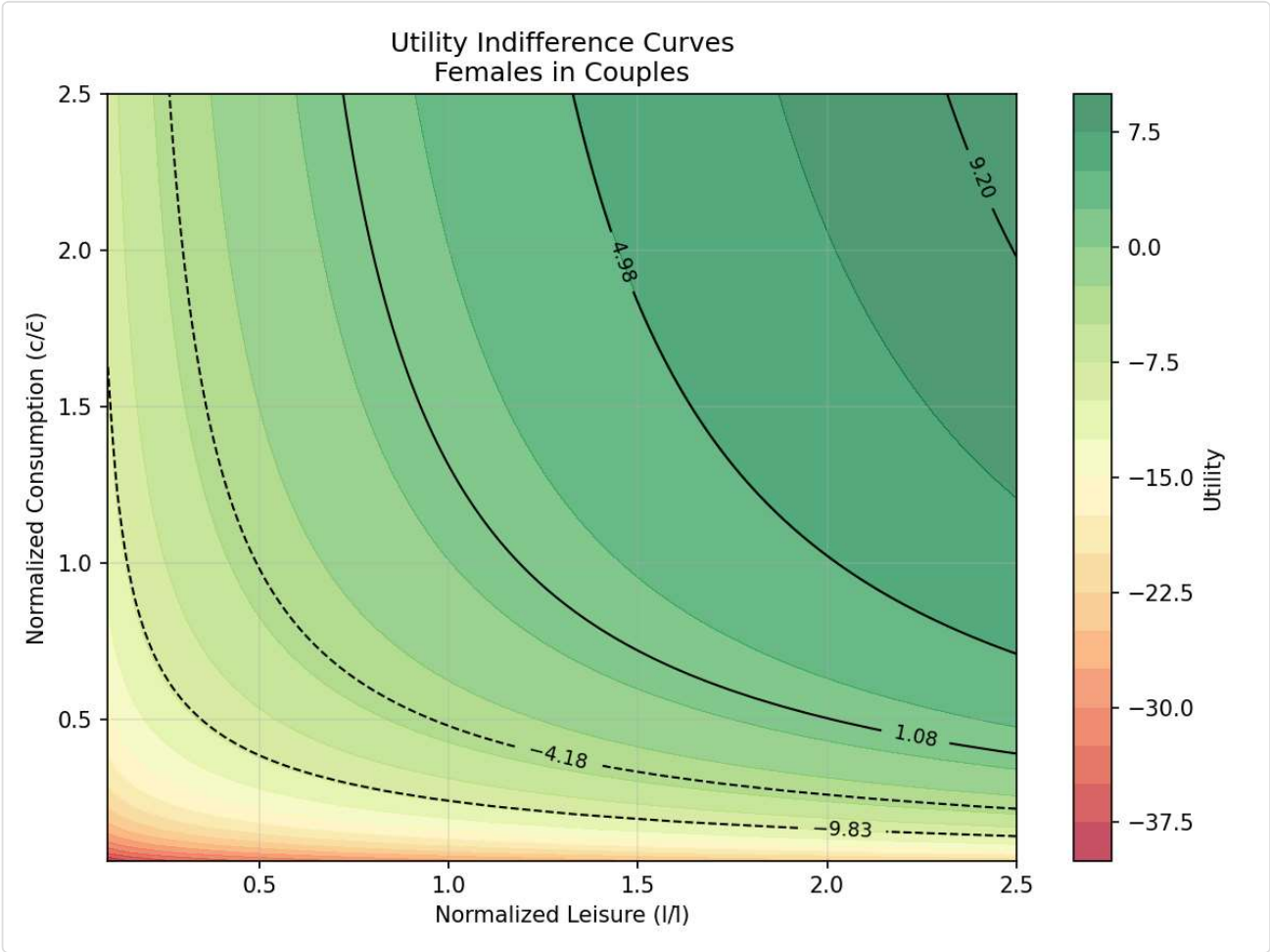
Single Females



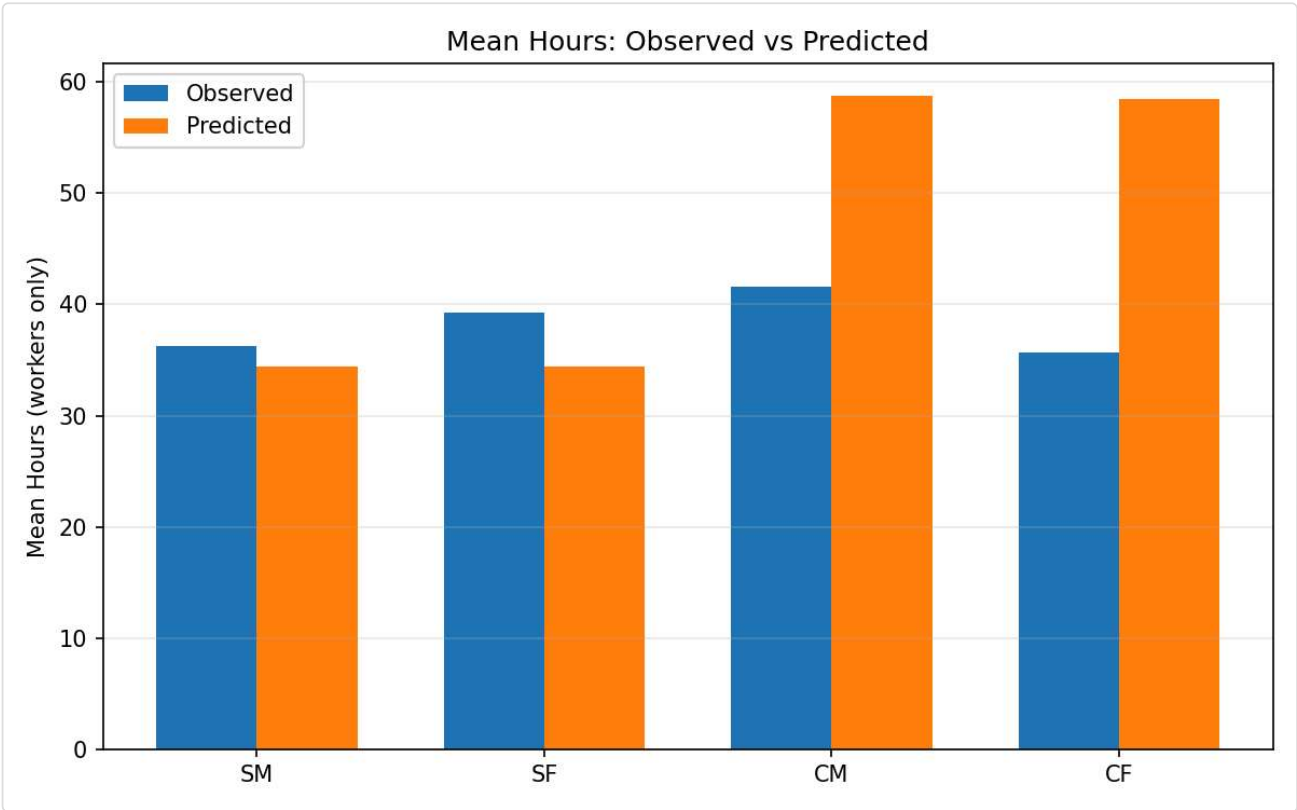
Males in Couples



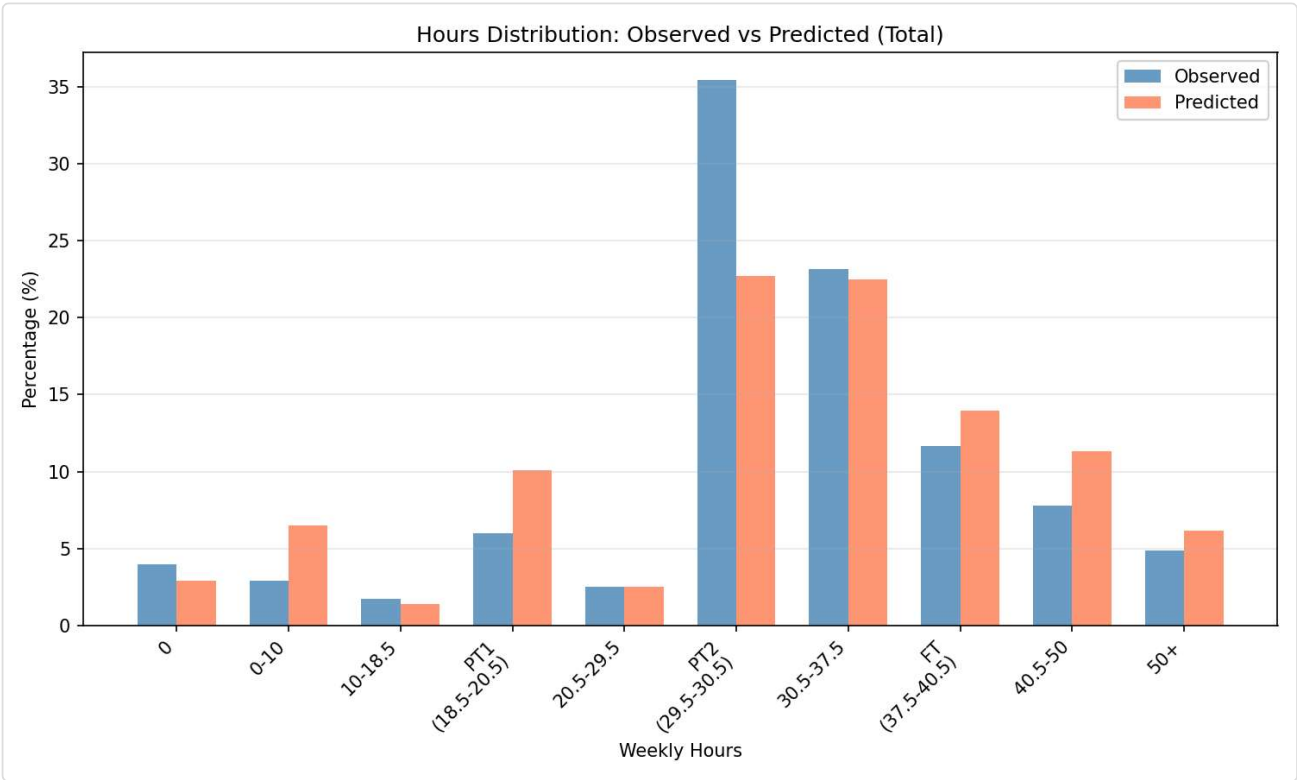
Females in Couples

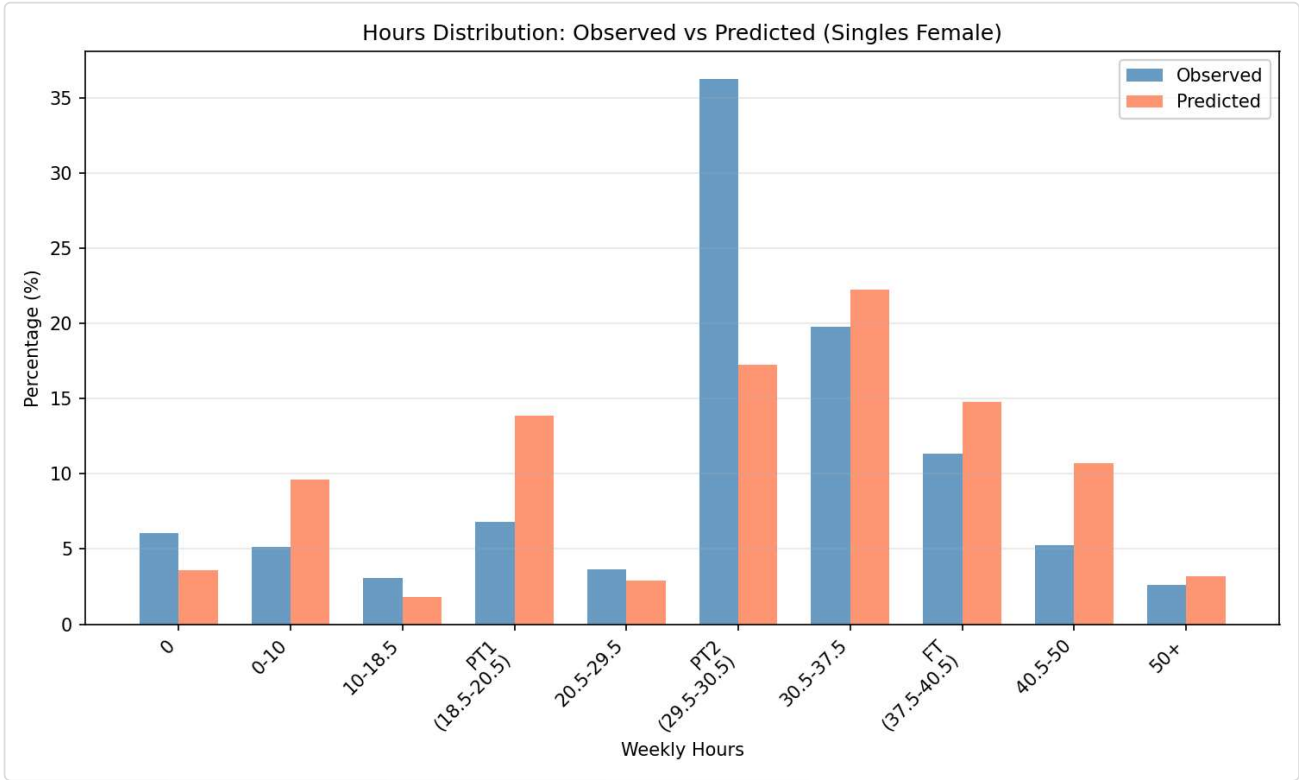
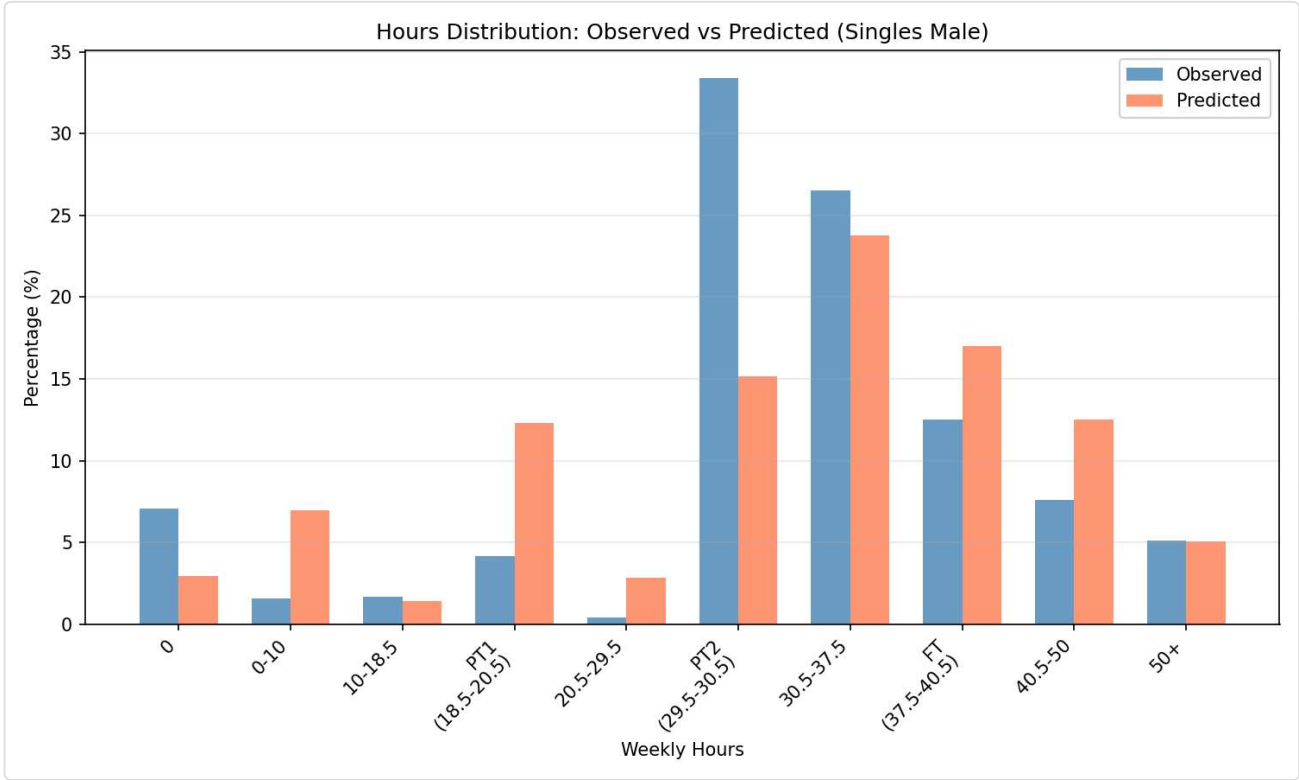


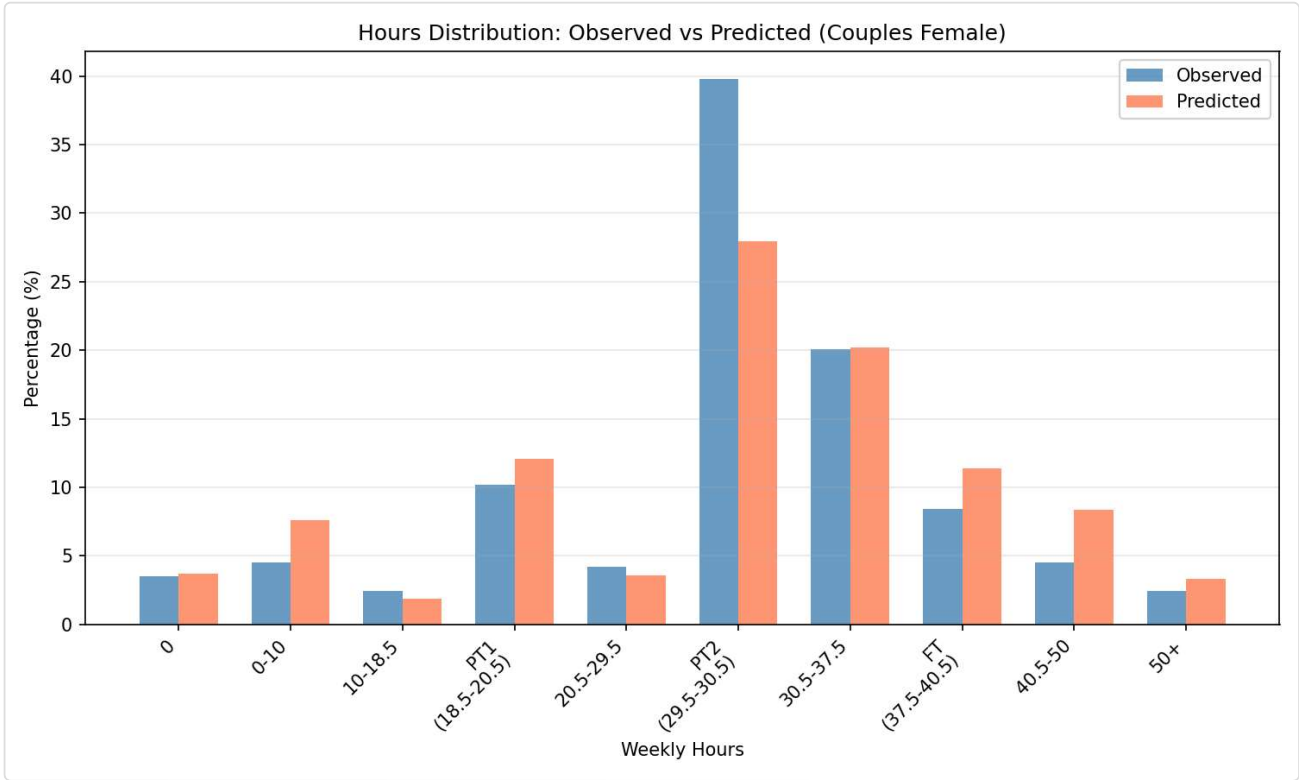
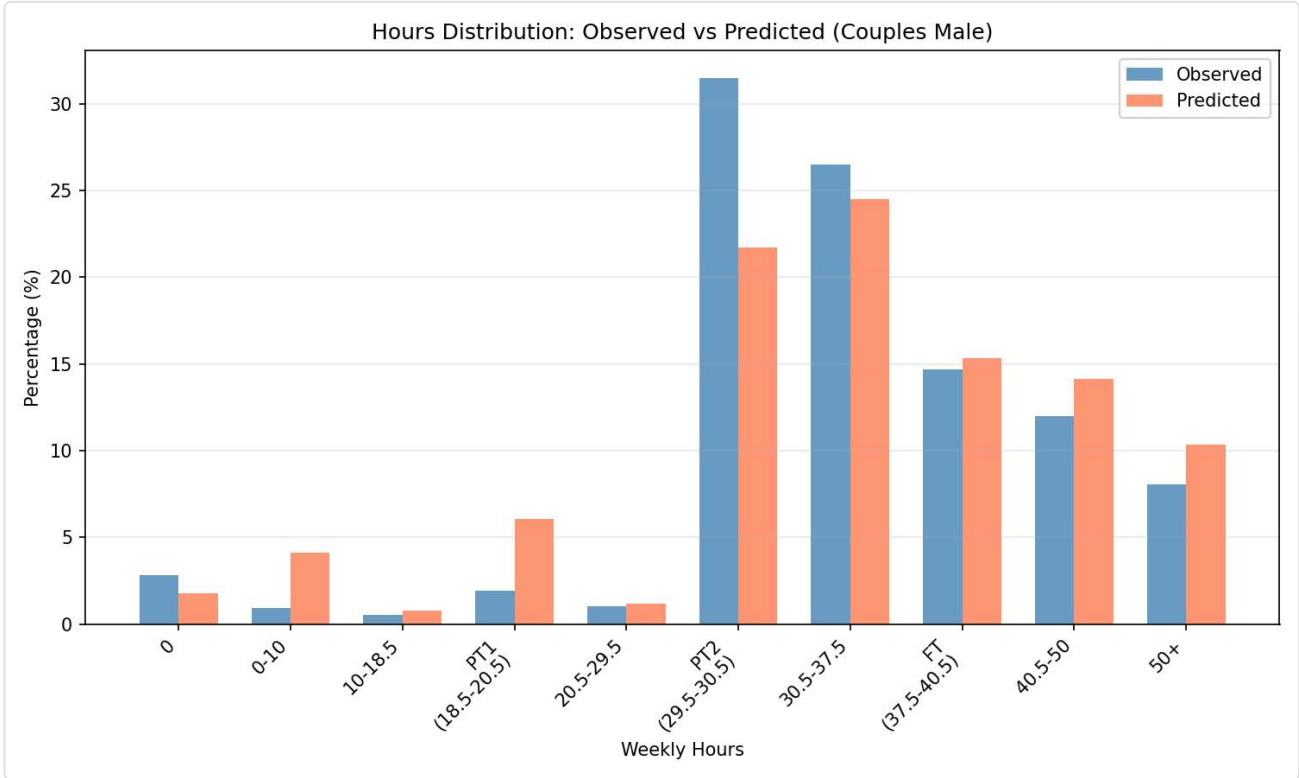
Mean Hours



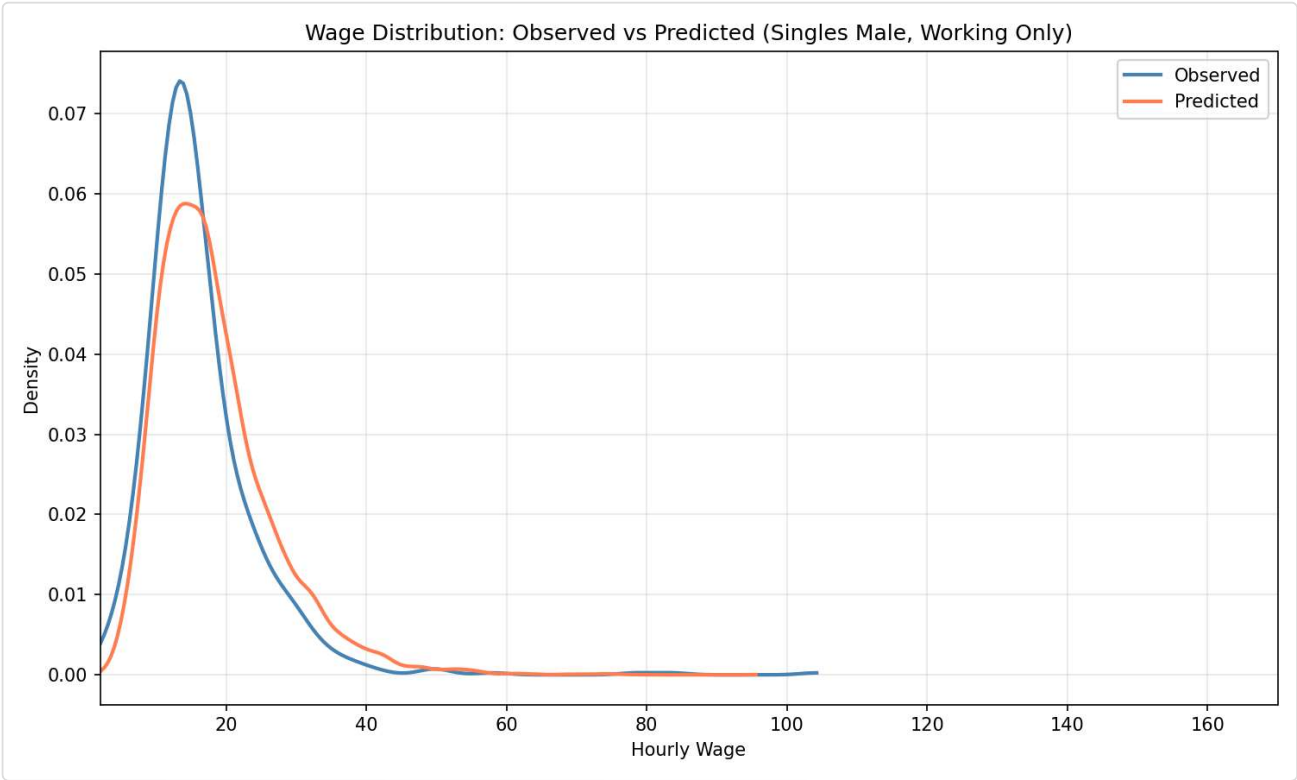
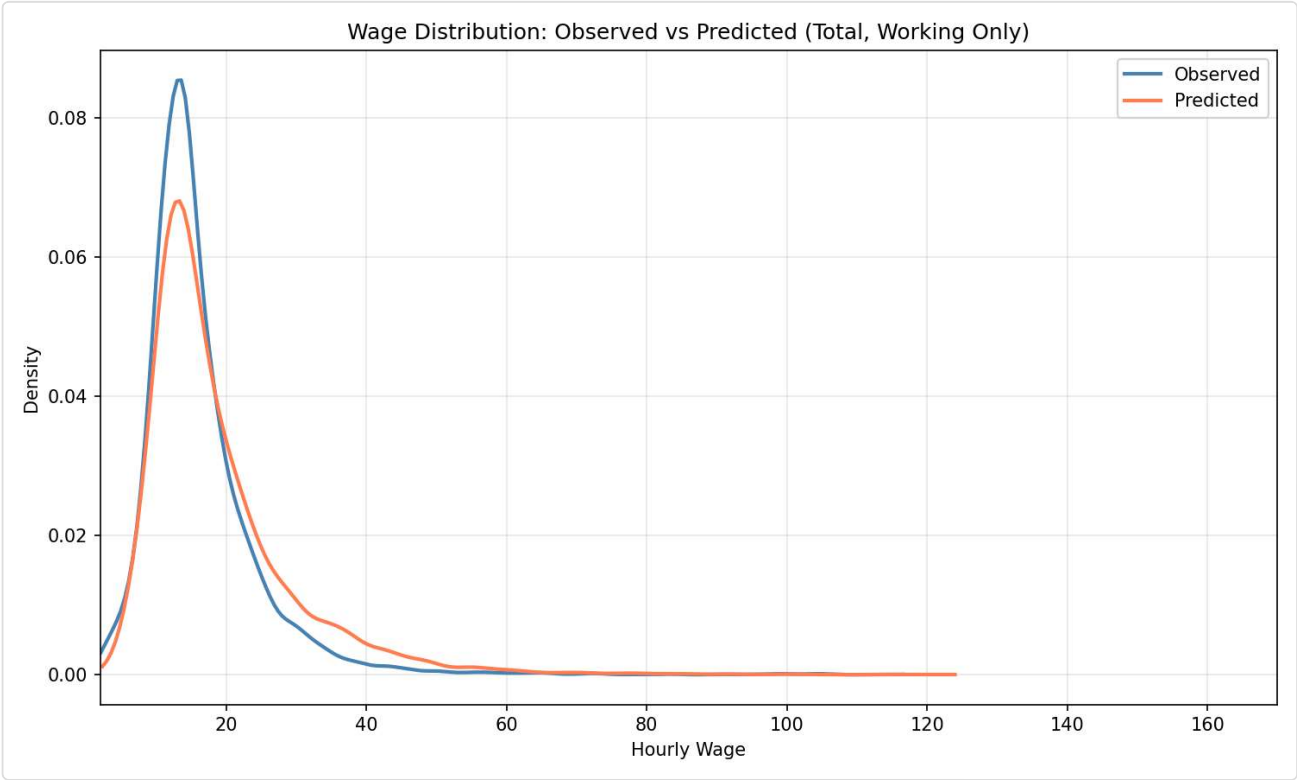
?? Hours Distribution: Observed vs Predicted

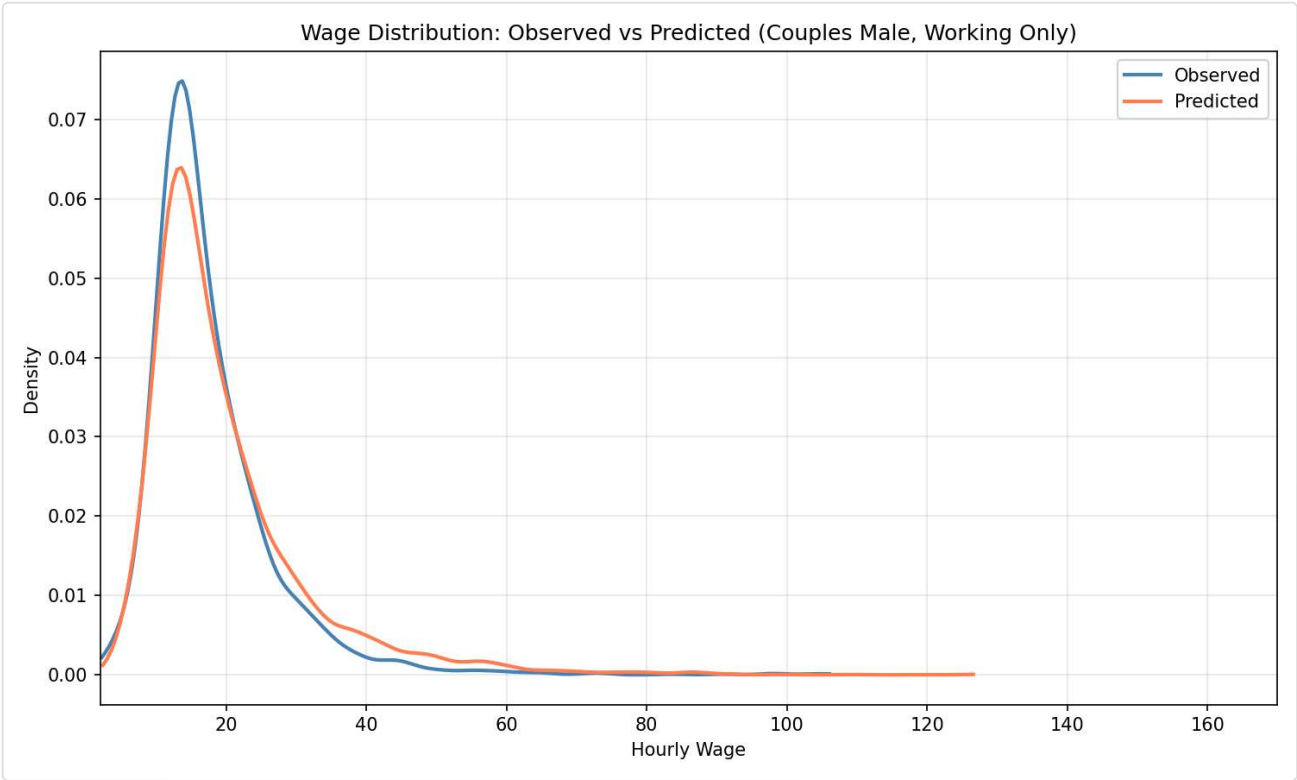
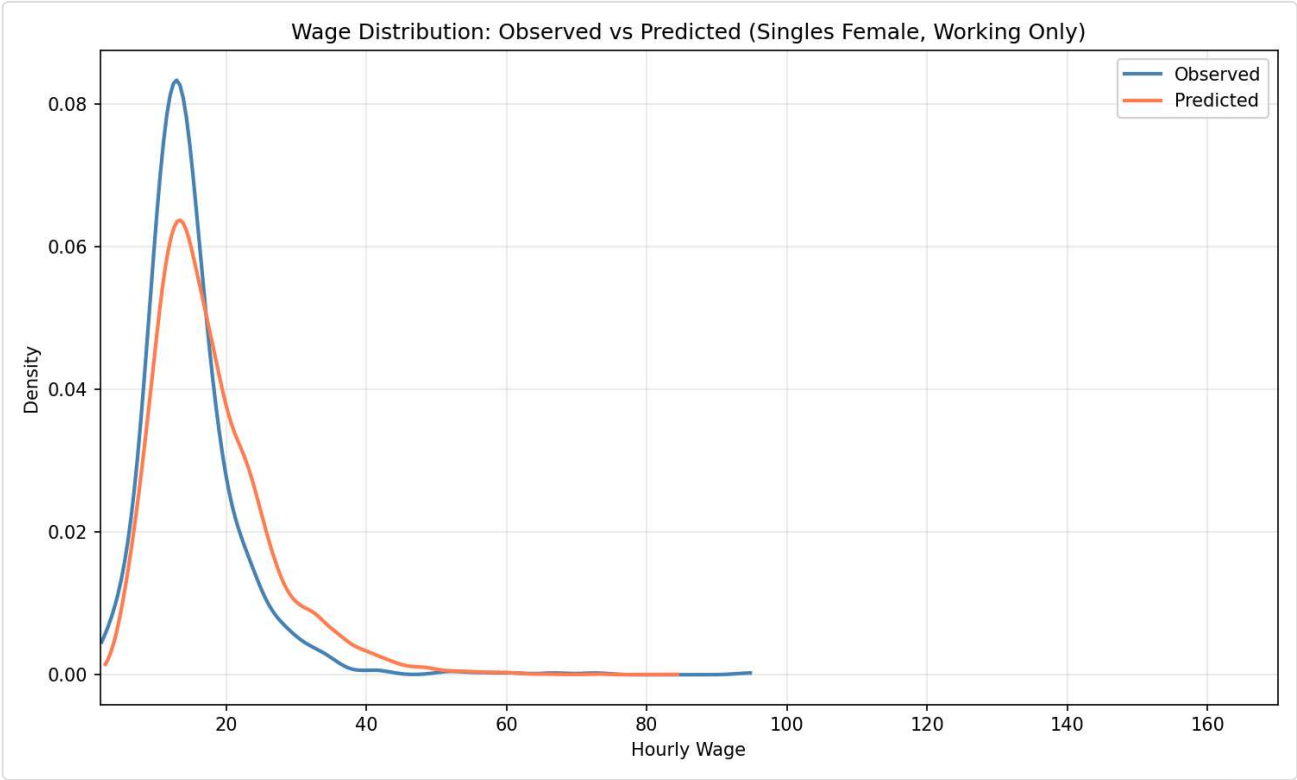


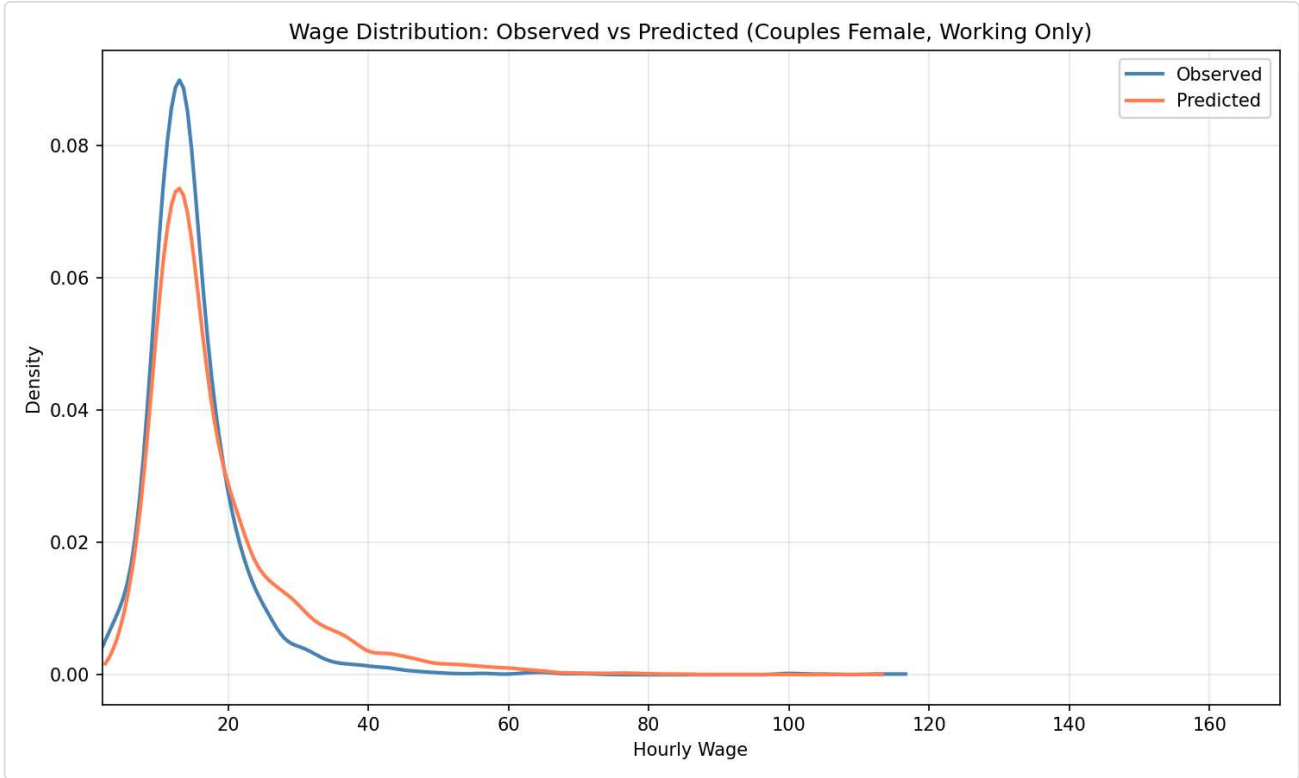




?? Wage Distribution: Observed vs Predicted (Working Only)







Group-Specific Parameters

Single Males

beta_E	-2.7597
beta_E_educH	0.5495
beta_E_gsur	-0.7408
beta_c	0.7475
beta_h_ft	1.4536
beta_h_pt1	-0.4991
beta_h_pt2	0.3772
beta_l0	3.9530
beta_l_age	0.0099
beta_l_age2	0.0020
beta_occ_2	-1.5154

Single Females

beta_E	-2.7597
beta_E_educH	0.5495
beta_E_gsur	-0.7408
beta_c	0.6767
beta_h_ft	1.4536
beta_h_pt1	-0.4991
beta_h_pt2	0.3772
beta_l0	4.5315
beta_l_age	0.0027
beta_l_age2	0.0042
beta_l_nkids	0.0949

beta_occ_2_cf	0.1878
beta_occ_2_cm	-1.4730
beta_occ_3	-2.1726
beta_occ_3_cf	-0.2020
beta_occ_3_cm	-2.2112
beta_occ_4	0.0169
beta_occ_4_cf	1.1237
beta_occ_4_cm	0.4791
beta_w0	2.0476
beta_w_educH	0.3007
beta_w_educL	-0.0478
beta_w_pexp	0.0167
beta_w_pexp2	-0.0002
sigma	0.4189
theta_c	-0.8357
theta_l	-0.7073

beta_occ_2	-0.0211
beta_occ_2_cf	0.1878
beta_occ_2_cm	-1.4730
beta_occ_3	-0.5724
beta_occ_3_cf	-0.2020
beta_occ_3_cm	-2.2112
beta_occ_4	0.7876
beta_occ_4_cf	1.1237
beta_occ_4_cm	0.4791
beta_w0	2.0476
beta_w_educH	0.3007
beta_w_educL	-0.0478
beta_w_pexp	0.0167
beta_w_pexp2	-0.0002
sigma	0.4189
theta_c	-0.8357
theta_l	-0.7268

Males in Couples

beta_E	-2.7597
beta_E_educH	0.5495
beta_E_gsur	-0.7408
beta_c	6.1537
beta_h_ft	1.4536
beta_h_pt1	-0.4991
beta_h_pt2	0.3772
beta_l0	2.3263
beta_l_age	-0.0044
beta_l_age2	0.0015
beta_occ_2_cf	0.1878
beta_occ_2_cm	-1.4730

Females in Couples

beta_E	-2.7597
beta_E_educH	0.5495
beta_E_gsur	-0.7408
beta_c	6.1537
beta_h_ft	1.4536
beta_h_pt1	-0.4991
beta_h_pt2	0.3772
beta_l0	4.8450
beta_l_age	-0.0467
beta_l_age2	0.0024
beta_l_nkids	0.2412
beta_occ_2_cf	0.1878

beta_occ_3_cf	-0.2020
beta_occ_3_cm	-2.2112
beta_occ_4_cf	1.1237
beta_occ_4_cm	0.4791
beta_w0	2.0476
beta_w_educH	0.3007
beta_w_educL	-0.0478
beta_w_pexp	0.0167
beta_w_pexp2	-0.0002
sigma	0.4189
theta_c	0.3189
theta_l	-0.7392

beta_occ_2_cm	-1.4730
beta_occ_3_cf	-0.2020
beta_occ_3_cm	-2.2112
beta_occ_4_cf	1.1237
beta_occ_4_cm	0.4791
beta_w0	2.0476
beta_w_educH	0.3007
beta_w_educL	-0.0478
beta_w_pexp	0.0167
beta_w_pexp2	-0.0002
sigma	0.4189
theta_c	0.3189
theta_l	-0.6970

 Parameter Estimates by Category

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

- ☐ Bounded (has constraints)
- ☐  Hit bound
- ☐ Warning: degenerate SE (near zero)
- ☐ $p \in [0.05, 0.1)$
- ☐ $p \in [0.1, 0.25)$
- ☐ $p \geq 0.25$

 Preference Parameters (Utility Function)

Parameters determining the utility from consumption (β_c, θ_c) and leisure (β_l, θ_l) by demographic group.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l0_sm	3.9530	0.7111	5.56	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sm	0.0099	0.0249	0.40	0.6913	-5.0000	5.0000	0.0000
joint.beta_l_age2_sm	0.0020	0.0021	0.96	0.3373	-1.0000	1.0000	0.0000
joint.beta_c_sm	0.7475	0.1499	4.99	0.0000 ***	0.0500	50.0000	1.0000
joint.theta_l_sm	-0.7073	0.1514	-4.67	0.0000 ***	-8.0000	0.9500	-1.0000

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.theta_c_sm	-0.8357	0.1428	-5.85	0.0000 ***	-8.0000	0.9500	-1.0000
joint.beta_l0_sf	4.5315	0.2673	16.95	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sf	0.0027	0.0271	0.10	0.9203	-5.0000	5.0000	0.0000
joint.beta_l_age2_sf	0.0042	0.0026	1.64	0.1018	-1.0000	1.0000	0.0000

 Employment and Hours Opportunity Parameters

Working-gated employment intercept and hours focal-point indicators, plus residual market-access shifters (gsur, education) declared in `market_opportunity.shifters`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_E	-2.7597	0.2985	-9.24	0.0000 ***	-25.0000	25.0000	0.0000
joint.beta_h_pt1	-0.4991	0.1077	-4.63	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_h_pt2	0.3772	0.1111	3.40	0.0007 ***	-10.0000	10.0000	0.0000
joint.beta_h_ft	1.4536	0.0500	29.10	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_E_gsur	-0.7408	0.2194	-3.38	0.0007 ***	-10.0000	10.0000	0.0000
joint.beta_E_educH	0.5495	0.2352	2.34	0.0195 *	-10.0000	10.0000	0.0000

 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\mu_w(X)$ intercept, education, experience, and residual standard deviation σ .

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_w0	2.0476	0.0255	80.33	0.0000 ***	-10.0000	20.0000	2.0000
joint.beta_w_educL	-0.0478	0.0204	-2.35	0.0190 *	-5.0000	5.0000	-0.1000
joint.beta_w_educH	0.3007	0.0145	20.75	0.0000 ***	-5.0000	5.0000	0.2000
joint.beta_w_pexp	0.0167	0.0022	7.69	0.0000 ***	-1.0000	1.0000	0.0200
joint.beta_w_pexp2	-0.0002	0.0000	-4.05	0.0001 ***	-0.1000	0.1000	-0.0003
joint.sigma	0.4189	0.0033	128.79	0.0000 ***	0.1000	20.0000	0.5000

 Occupation Opportunity Parameters

Working-gated occupation shifters declared in `occupation_opportunity.shifters`; routed by `applies_to` group (sm/sf/cm/cf). Reference category is the YAML's `occupation_opportunity.reference`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_2_sm	-1.5154	0.1424	-10.64	0.0000 ***	-10.0000	10.0000	0.0000

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_3_sm	-2.1726	0.1847	-11.76	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sm	0.0169	0.0862	0.20	0.8445	-10.0000	10.0000	0.0000
joint.beta_occ_2_sf	-0.0211	0.1129	-0.19	0.8521	-10.0000	10.0000	0.0000
joint.beta_occ_3_sf	-0.5724	0.1292	-4.43	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sf	0.7876	0.0922	8.55	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cm	-1.4730	0.1134	-12.99	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_cm	-2.2112	0.1473	-15.02	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_cm	0.4791	0.0685	7.00	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_sf	0.1878	0.0888	2.11	0.0368 *	-10.0000	10.0000	0.0000



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